

James Queeney

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RESEARCH SKILLS

I am interested in developing reliable, data-driven methods for decision making and control. My current research focuses on the need for robustness, safety, and generalization in deep reinforcement learning, imitation learning, and self-supervised learning, with applications in robotics.

- **Research Areas:** deep reinforcement learning, imitation learning, self-supervised learning, robust data-driven optimization and control, uncertainty quantification, robotics
- **Programming Languages:** Python, MATLAB, R
- **Software Experience:** Gurobi, Isaac Lab, MuJoCo, PyTorch, TensorFlow
- **Hardware Experience:** sim-to-real policy transfer on Unitree Go2 quadruped robot

EDUCATION

Boston University Aug 2023

PhD in Systems Engineering

- Dissertation: “Reliable deep reinforcement learning: Stable training and robust deployment”

Boston University Jan 2022

MS in Systems Engineering

Colgate University May 2013

BA in Mathematics and Mathematical Economics

- Class of 2013 Valedictorian, Honors in Mathematics, Phi Beta Kappa, Summa Cum Laude

RESEARCH EXPERIENCE

Research Scientist 2025 – Present

Amazon Robotics

- Topic: Optimization, planning, and task assignment for robotic systems

Postdoctoral Research Fellow 2023 – 2025

Mitsubishi Electric Research Laboratories

- Topic: Robustness, safety, and generalization in learning-based robotic control

Research Affiliate 2023 – 2025

Massachusetts Institute of Technology – Host: Jonathan How

- Topic: Uncertainty-aware learning for robust planning and control of mobile robots

Doctoral Research Fellow 2019 – 2023

Boston University – Advisors: Ioannis Paschalidis, Christos Cassandras

- Topic: Reliable deep reinforcement learning with performance guarantees

Research Intern Summer 2022

Mitsubishi Electric Research Laboratories – Host: Mouhacine Benosman

- Publication: “Risk-averse model uncertainty for distributionally robust safe reinforcement learning”

Research Assistant 2017 – 2018
Colgate University – Host: William Cipolli
• Topic: Bayesian non-parametric approaches to supervised learning with Polya trees

INDUSTRY EXPERIENCE

Director of Operations Research 2017 – 2018
Bargain Hunt

Private Equity Associate 2015 – 2017
Thomas H. Lee Partners – Consumer & Healthcare Group

Investment Banking Analyst 2013 – 2015
Bank of America Merrill Lynch – Mergers & Acquisitions Group

TEACHING AND OUTREACH

CISE Graduate Student Workshop Organizer Jan 2023
Boston University Center for Information & Systems Engineering

Graduate Teaching Fellow Fall 2022
Boston University – Optimization Theory and Methods (SE 674)

Research Mentor Summer 2021
Boston University Research in Science & Engineering Program

Graduate Teaching Fellow Fall 2019
Boston University – Introduction to Programming for Engineers (EK 125)

PUBLICATIONS

Preprints

- Queeney, J., Cai, X., Schperberg, A., Corcodel, R., Benosman, M., and How, J. P. (2025). GRAM: Generalization in deep RL with a robust adaptation module. arXiv:2412.04323.
- Chen, Y., Giammarino, V., Queeney, J., and Paschalidis, I. C. (2024). Provably efficient off-policy adversarial imitation learning with convergence guarantees. arXiv:2405.16668.

Peer-Reviewed Publications

- Giammarino, V., Queeney, J., and Paschalidis, I. C. (2025). Visually robust adversarial imitation learning from videos with contrastive learning. In *IEEE International Conference on Robotics and Automation (ICRA 2025)*.
- Cai, X., Queeney, J., Xu, T., Datar, A., Pan, C., Miller, M., Flather, A., Osteen, P. R., Roy, N., Xiao, X., and How, J. P. (2025). PIETRA: Physics-informed evidential learning for traversing out-of-distribution terrain. *IEEE Robotics and Automation Letters (RA-L)*.
- Queeney, J., Paschalidis, I. C., and Cassandras, C. G. (2025). Generalized policy improvement algorithms with theoretically supported sample reuse. *IEEE Transactions on Automatic Control (TAC)*.
- Ozcan, E. C., Giammarino, V., Queeney, J., and Paschalidis, I. C. (2024). A model-based approach for improving reinforcement learning efficiency leveraging expert observations. In *63rd IEEE Conference on Decision and Control (CDC 2024)*.
- Giammarino, V., Queeney, J., and Paschalidis, I. C. (2024). Adversarial imitation learning from visual observations using latent information. *Transactions on Machine Learning Research (TMLR)*.

- Queeney, J., Ozcan, E. C., Paschalidis, I. C., and Cassandras, C. G. (2024). Optimal transport perturbations for safe reinforcement learning with robustness guarantees. *Transactions on Machine Learning Research (TMLR)*.
- Queeney, J. and Benosman, M. (2023). Risk-averse model uncertainty for distributionally robust safe reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS 2023)*.
- Giammarino, V., Queeney, J., Carstensen, L. C., Hasselmo, M. E., and Paschalidis, I. C. (2023). Opportunities and challenges from using animal videos in reinforcement learning for navigation. In *The 22nd World Congress of the International Federation of Automatic Control (IFAC 2023)*.
- Queeney, J., Paschalidis, I. C., and Cassandras, C. G. (2021). Generalized proximal policy optimization with sample reuse. In *Advances in Neural Information Processing Systems (NeurIPS 2021)*.
- Queeney, J., Paschalidis, I. C., and Cassandras, C. G. (2021). Uncertainty-aware policy optimization: A robust, adaptive trust region approach. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI 2021)*.

Dissertation

- Queeney, J. (2023). *Reliable deep reinforcement learning: Stable training and robust deployment*. PhD thesis, Boston University.

PRESENTATIONS

- Reliable deep reinforcement learning for robotics (2025). *Amazon Robotics Invited Talk*, Virtual.
- Risk-averse model uncertainty for distributionally robust safe reinforcement learning (2023). *37th Conference on Neural Information Processing Systems (NeurIPS 2023)*, New Orleans, LA.
- Reliable deep reinforcement learning: Stable training and robust deployment (2023). *Boston University Division of Systems Engineering PhD Final Defense*, Boston, MA.
- Reliable deep reinforcement learning with robustness and safety guarantees (2023). *Mitsubishi Electric Research Laboratories Invited Talk*, Cambridge, MA.
- Safe reinforcement learning with robustness guarantees (2023). *Massachusetts Institute of Technology Invited Talk*, Cambridge, MA.
- Balancing stability and efficiency in deep reinforcement learning (2023). *Harvard University Invited Talk*, Cambridge, MA.
- Stable and efficient reinforcement learning with principled sample reuse (2022). *CISE Graduate Student Workshop 8.0*, Boston, MA. **Best Presenter Award**.
- Robust and efficient reinforcement learning from limited data (2021). *Boston University Division of Systems Engineering PhD Prospectus Defense*, Boston, MA.
- Generalized proximal policy optimization with sample reuse (2021). *35th Conference on Neural Information Processing Systems (NeurIPS 2021)*, Virtual.
- Uncertainty-aware policy optimization: A robust, adaptive trust region approach (2021). *CISE Best Student Paper Awards Presentation*, Virtual. **Best Student Paper Award Finalist**.
- Uncertainty-aware policy optimization: A robust, adaptive trust region approach (2021). *35th AAAI Conference on Artificial Intelligence (AAAI 2021)*, Virtual.

HONORS AND AWARDS

• Doctoral Research Fellow, <i>Boston University</i>	2019 – 2023
• CISE Best Student Paper Award Finalist, <i>Boston University</i>	2022
• CISE Graduate Student Workshop Best Presenter Award, <i>Boston University</i>	2022
• CISE Best Student Paper Award Finalist, <i>Boston University</i>	2021
• Dean's Fellowship Award, <i>Boston University</i>	2018 – 2019
• Class of 2013 Valedictorian, <i>Colgate University</i>	2013
• Osborne Mathematics Prize, <i>Colgate University</i>	2013
• Phi Beta Kappa Award, <i>Colgate University</i>	2013
• Phi Beta Kappa, <i>Colgate University</i>	2013
• Summa Cum Laude, <i>Colgate University</i>	2013
• Honors in Mathematics, <i>Colgate University</i>	2013
• John T. Mitchell Award, <i>Colgate University</i>	2012 – 2013
• Charles A. Dana Scholar, <i>Colgate University</i>	2011 – 2013
• Alumni Memorial Scholar, <i>Colgate University</i>	2009 – 2013
• Sisson Mathematics Prize, <i>Colgate University</i>	2010
• Dodge Prize, <i>Colgate University</i>	2010